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Temporal evolution of the hydromechanical properties of soil-root systems in a forest fire in China

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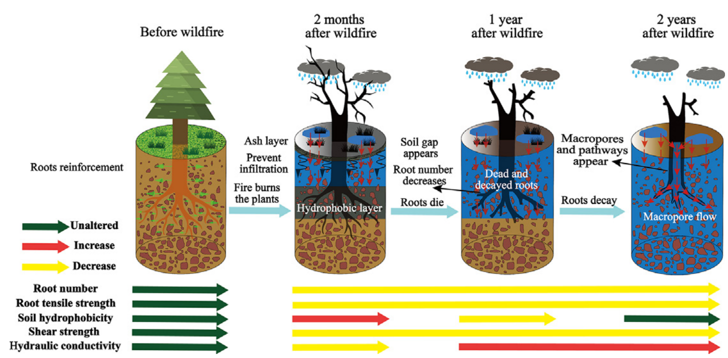
HIGHLIGHTS

- Wildfire reduced both root number and tensile strength, decreasing soil strength.
- After the wildfire, hydrophobicity and ash clogging decreased hydraulic conductivity.
- One year after fire, macropore formation induced hydraulic conductivity recovery.

GRAPHICAL ABSTRACT



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ABSTRACT

Plant roots generally enhance soil strength and stabilize slopes through hydro-mechanical effects, especially in forested areas prone to shallow slope failure. Forest fires can severely weaken the hydro-mechanical contribution of roots to slopes, however, the hydro-mechanical characteristics of soil–root systems (SRS) affected by wildfire remain poorly understood. To obtain insight into the post-fire hydro-mechanical characteristics of SRS, a subalpine conifer forested area in Sichuan Province, China that suffered a wildfire on March 30, 2019 was continuously monitored over two consecutive years. Samples from zones with different degrees of burn severity were collected and tests both for roots and SRS were performed. The results revealed a substantial decline in root number, which decreased by 46%–58% two years after the wildfire in the medium- and high-severity areas. The tensile strength tests indicated a reduction of root tensile strength by 36%–47% for roots with diameters less than 2 mm. The shear strength of the SRS determined from saturated direct shear tests strongly and had degraded by 55%–82% two years after the wildfire because of root death and reduced root reinforcement. The results of hydraulic conductivity tests over the same time period indicated an abrupt reduction of SRS hydraulic conductivity within several months after the fire owing to ash clogging and the formation of a hydrophobic layer. After more time had elapsed, however, hydraulic conductivity had increased unexpectedly by a factor of 2.2–3.2 greater than that of unburned soil. We attribute this observation to the formation of macropore flow pathways from decayed roots, which was observed by scanning electron microscopy. The findings presented here provide important insight into the temporal changes of the hydro-mechanical characteristics of SRS in burned areas and their associated mechanisms and could be a useful reference to better evaluate post-wildfire stability of subalpine conifer forest in similar environmental conditions.

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1. Introduction

Forest fire is a destructive natural disaster that can combust vegetation and destroy vegetation cover, alter soils infiltration, decrease water quality, and lead to water-induced hazards (Certini et al., 2021; Certini, 2005; Jalaludin et al., 2020; Lucas-Borja et al., 2020b). Previous studies have shown that secondary geological disasters, such as debris flows and shallow landslides (Cui et al., 2021), occur in approximately 40% of burned areas (Hu et al., 2018). Worldwide, these have occurred frequently in recent years. For example, on January 9, 2018, a post-fire debris flow in California, United States caused 23 fatalities, injured at least 167 people, and damaged 408 houses (Kean et al., 2019). On June 20, 2018, a post-fire debris flow caused by runoff in the West Cascade Range, Oregon, United States caused road damage and traffic jams (Wall et al., 2020). On June 20, 2016, the San Gabriel Complex forest fire broke out in southern California, United States. This fire left a total burned area of 21 km^2 and led to widespread shallow landslides in the years following the wildfire (Rengers et al., 2020).

The occurrence of post-fire debris flows and shallow landslides is closely related to the status of the soil after the fire (Bodí et al., 2012; Cui et al., 2015; Cui et al., 2019; Santín et al., 2018). Previous studies have mostly investigated the physical, chemical, and biological properties of soil shortly after a fire (DeBano, 2000; González-Pérez et al., 2004; Lucas-Borja et al., 2020b; Rodríguez et al., 2017; Rodríguez et al., 2018). To date, however, little attention has been paid to the changes in the hydro-mechanical properties of soil–root systems (SRS), which are directly related to slope stability (Inbar et al., 2014; López-Martín et al., 2018; Neary et al., 2005). Roots can reinforce soil and increase its strength (Arnone et al., 2016; Giadrossich et al., 2019), and increase the matric suction of soil by absorbing water (Ng et al., 2016a; Ng et al., 2016b). In addition, root exudates can also enhance the agglomeration of soil particles (Daynes et al., 2013) and increase soil erosion resistance (Yao et al., 2020). The high temperatures of fires can lead to root death, which may ultimately alter the mechanical and hydrological functions of SRS (Lei et al., 2021; Ni et al., 2018; Stokes et al., 2009). However, the temporal evolution of the mechanical and hydrological properties of SRS after wildfires remains unclear, which limits our understanding of the initiation mechanisms of post-fire debris flows and shallow landslides.

The objectives of this study are to (1) investigate the changes of root and SRS characteristics in areas that have experienced variable fire severity as a function of post-fire time and (2) explore the causes of short-term hydraulic and mechanical evolution from an experimental and microscopic perspective to better understand the initiation mechanisms of post-fire debris flows and shallow landslides. An area burned by the March 30, 2019 forest fire in Sichuan Province, China was selected for continuous monitoring for two years. Field and laboratory analysis were performed to determine parameters including number and tensile strength of roots, water drop penetration time (WDPT), shear strength and hydraulic conductivity of SRS as well as the microstructure characteristics of SRS.

2. Materials and methods

2.1. Study area

The study area is located in Muli County, Liangshan Prefecture, Sichuan Province, China. It is in the middle of undulating terrain in the Hengduan Mountains and is characterized as a typical alpine canyon area (Fig. 1). Muli County contains one of the three major forest farms in China, with 67.3% forest coverage (Rao et al., 2019), where the main species of tree are *Picea likiangensis* (*Franch*) *Pritz*, *Picea brachytyla* (*Franch.*) *Pritz*, and *Pinus yunnanensis* (Li, 1993; Rao et al., 2019). The mean annual temperature and precipitation in study area are 12.0 °C and 854.6 mm, respectively. Rainfall occurs predominantly in summer and the wettest month is July, with a mean rainfall of 220.2 mm, while the driest month is generally January, with a mean rainfall of 1.7 mm. The highest temperature in the whole year is concentrated in July, with a monthly average temperature of 17.5 °C, and the coldest month is January, with a monthly average temperature of 4.7 °C (Liu, 2009). In addition, the site elevations ranged from 2865 m to 3719 m, with generally steep slopes (0–65.1°). At 18:00 on March 30, 2019, a serious forest fire occurred in Lier village (28°31′50.20″N, 101°13′34.21″ E), Yalongjiang town, Muli County, leaving a burned area of approximately 20 ha and causing 31 deaths (BJNEWS, 2019).

The main forest tree species in the burned area prior to the wildfire was *Pinus yunnanensis*, which formed a commercial forest land, with an average stem diameter of 8–20 cm (DBH). The understory of *Pinus yunnanensis* forest was composed mainly of *Rhododendron rupicola* var. *muliense*, *Dodonaea viscosa* (L.) *Jacq*, *Ageratina adenophora*, and other shrubs and herbs. Near the unburned area, we established six square quadrats with side length of 30 m to investigate the forest stand characteristics. The stem diameter at breast height was 17.4 ± 2.7 cm in the plots, whereas the mean stand height, stand density, and basal area were 13.5 ± 1.8 m, 1496 ± 63 trees/ha and 37.2 ± 3.5 m^2/ha, respectively. The soil, which was dominated by brown soil with a granulometric composition of loams (USDA, 2004), had developed on shale bedrock. The burn severity reflects the range of burning and degree of damage to the forest vegetation, which has an indirect impact on root decline after a fire and controls the vegetation recovery process (Lucas-Borja et al., 2019; Pereira et al., 2012). Two Sentinel-2 images collected on February 4, 2019 and April 20, 2019 were used to estimate the burn severity using the differenced normalized burn ratio (dNBR) method (Key and Benson, 2006). The near-infrared (resolution 10 m) band and short-wave infrared (resolution 20 m) band from the two pre-processed images were resampled into the same 20-m resolution. A difference map was then derived from the two NBR results. The dNBR range corresponding to each burn severity was calibrated according to the NBR change before and after the wildfire. Accompanied by the forest manager, we conducted a detailed investigation of the burned area and investigated the burning status of vegetation in the fire area according to the identification standard of burn severity (Table 1) (Gordillo-Rivero et al., 2014; Keeley, 2009) to check the burn severity. From this, a burn severity map of the site (Fig. 1c), low- (Fig. 1d), medium- (Fig. 1e), and high-severity (Fig. 1f) areas were distinguished, and sheet erosion (Fig. 1g), gully erosion (Fig. 1h), and shallow landslides (Fig. 1i) in high-severity areas could be observed.

2.2. Sample collection and field investigation

Samples of the roots and SRS in different burn severity zones were collected from the burned area two months, one year and two years after the fire. Samples were also taken from the unburned area as a reference. The root distribution for three tree stumps for each burn severity zone during each sampling period was investigated and sampled using the trench profile wall method (Bordoni et al., 2019; Ji et al., 2012). The sampled trees all had a diameter of approximately 15 cm, to minimize differences caused by tree size. Adjacent sampling locations were selected for the sampling of unburned (28°32′22″N, 101°16′14″E; 43.1° in slope, 3425 m in altitude), low- (28°32′27″N, 101°16′16″E; 41.7° in slope, 3309 m in altitude), medium- (28°32′24″N, 101°16′15″ E; 46.7° in slope, 3371 m in altitude) and high- (28°32′29″N, 101°16′ 13″E; 39.9° in slope, 3298 m in altitude) severity samples (Fig. 1c). On the one hand, this sampling method can make sure the consistency of soil properties and soil types at the sampling position before the fire (Leslie et al., 2014; Vergani et al., 2016); and on the other hand, this ensured that the slope and altitude were similar, which could reduce the impact of environmental factors on root distribution and properties (Robichaud et al., 2013b; Sainju and Good, 1993).

A ditch 1 m wide and 0.8 m deep was excavated at a radial distance of 0.6 m from each stump in the downslope direction to ensure that most of the roots would be exposed on the ditch surface. The

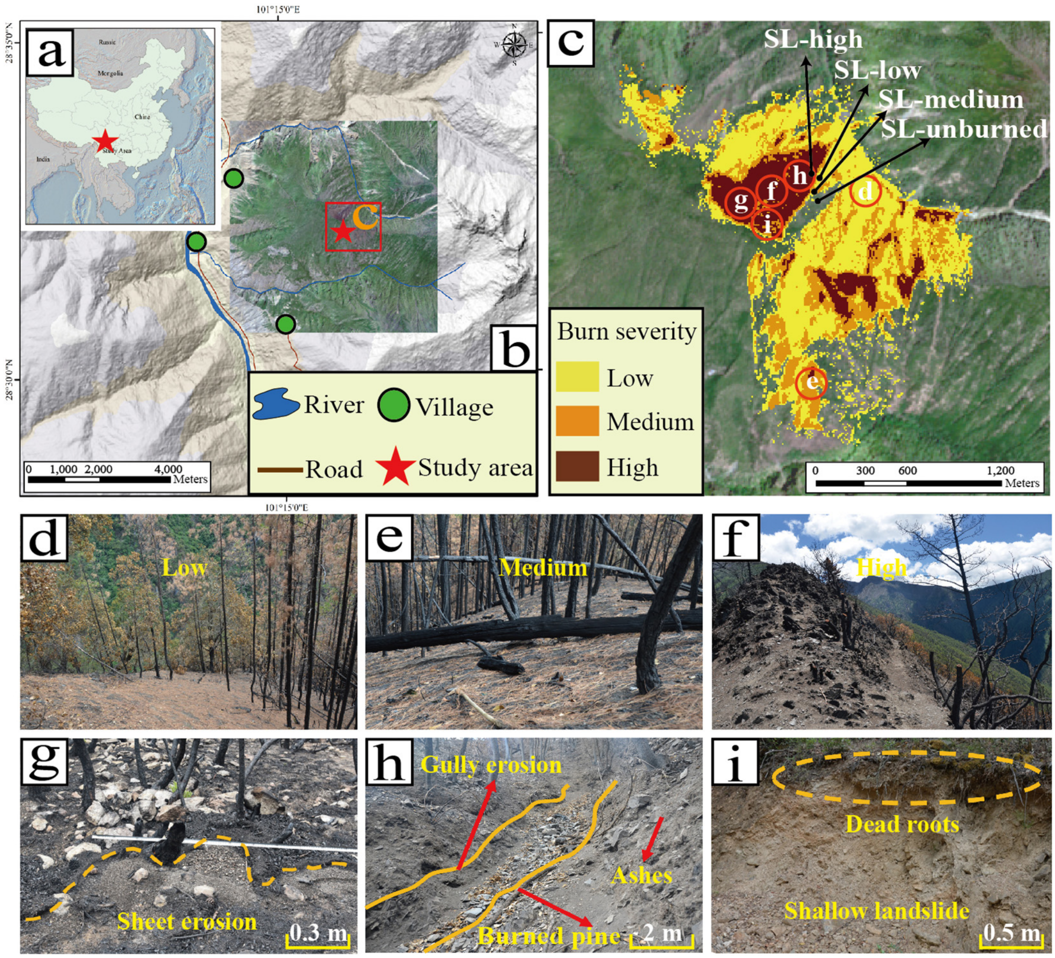

Fig. 1. Overview of the study area in Lier village, Yalongjiang town, Muli County, Sichuan Province, China. (a–b) Location of the study area; (c) burn severity map of the site, note. "SL-unburned", "SL-low", "SL-medium" and "SL-high" refer to sampling location from unburned, low-, medium-, and high-severity area, respectively; (d) photo of low-severity burned area; green needles of some pines were scorched; (e) photo of medium-severity burned area; pines and logs were charred; (f) photo of high-severity burned area; pines were burned, and white ash deposition and charred organic matter covered the slope; (g) sheet erosion; (h) gully erosion formed by post-fire debris flows; (i) shallow landslides caused by roots death.

distribution and number of pine tree roots were statistically investigated (Bischetti et al., 2009; Ji et al., 2012) for root diameters of 0.5–1, 1–2, 2–5, 5–10, and >10 mm. The roots were then preserved in a 15% alcohol solution to prevent deterioration (Vergani et al., 2014), and all experiments were completed within two weeks after sampling.

The soil samples of each burn severity were collected in a 5 × 10 m plot, where each plot was divided into 28 smaller circular sampling sites (Fig. 2), and the sampling location in the next year was the same as that in the previous year (Fig. 1c). After removing the charred organic layer from the surface, undisturbed surface SRS samples (0–5 cm in depth) were taken by a ring knife (diameter = 62 mm, height = 20 mm). Twenty-eight undisturbed SRS samples were collected within the plots (Fig. 2) in each sampling period for each burn severity area (a total of 84 samples from low-, medium- and high-severity areas). It was worth noting that this study focused on the impact of burn severity and post-fire time on the soil in the area disturbed by fire. Therefore, the basic assumption of this study was that the soil properties in the unburned area remained unchanged. Based on this assumption, twenty-eight samples in the control group from the unburned area were only collected at the time of the first sampling. In addition, remolded soil weighing about five kilograms from different burn severity conditions was also collected from the site. All the undisturbed and remolded soil samples collected in different burn severity areas and post-fire time, as well as samples of control group, were taken back to the laboratory for different laboratory tests (Table 2).

After sampling, WDPT tests were conducted near the sampling sites in the same plot (Fig. 2) (Jiménez-Pinilla et al., 2016) to evaluate the water repellency of the soil in different burn severity areas (Jiménez-Pinilla et al., 2016; Lichner et al., 2012). For the measurements, ten water drops were placed onto the soil 2 cm below the soil surface with a dropper bottle (Robichaud et al., 2013b). This was repeated in each burn severity zone during each sampling period. The WDPT was recorded when the water drop passed through the soil, and a mean WDPT was determined to categorize the soil water repellency of each site: "non-water repellency" (0–30 s), "weak water repellency"

Table 1
Identification standards for different burn severity areas (Gordillo-Rivero et al., 2014; Keeley, 2009).

Burn severity	Description
Unburned	Surface vegetation and trees are not affected by fire.
Low	Canopy trees have green needles, although stems are scorched; surface litter, mosses, and herbs charred or consumed.
Medium	Trees with some canopy cover killed and all understory plants charred or consumed; fine dead twigs on soil surface consumed and logs charred.
High	Canopy trees killed and needles consumed; surface litter of all sizes and soil organic layer largely consumed; white ash deposition and charred organic matter to depths of several cm.

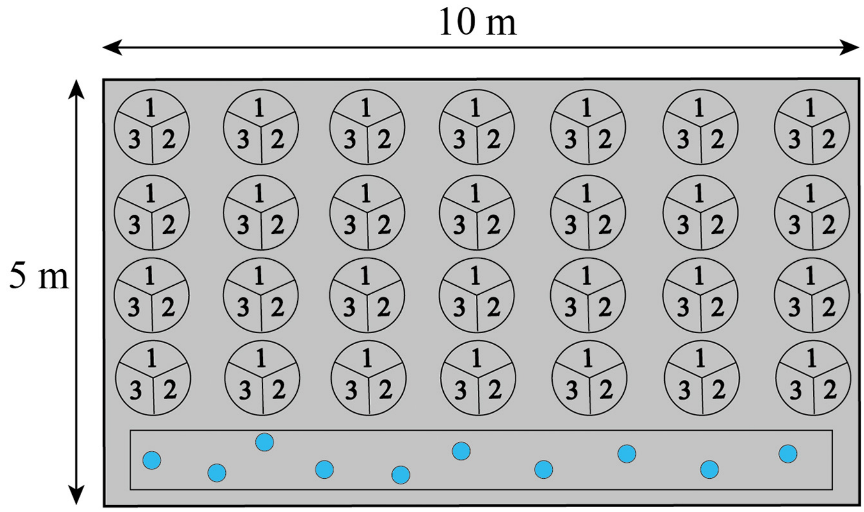

Fig. 2. Schematic diagram for sampling and WDPT tests in the field. Each plot has been divided into 28 circular circles with similar area approximately between them. The circle represents the sampling point, while the numbers in the circles represent each sampling effort. The rectangular wedge at the bottom of the plot is used for field WDPT test.

(31–60 s), "moderate water repellency" (61–180 s), and "strong water repellency" (181–360 s) (Dekker and Ritsema, 1994; Glenn and Finley, 2010).

2.3. Root tensile tests

Roots from unburned and low-, medium-, and high-severity areas were subjected to tensile tests two months, one year, and two years after the fire to explore the effects of burning on their mechanical properties. The root samples were sieved and straight root segments with a length of 10 cm were selected (Vergani and Graf, 2016). Tensile tests were carried out using a clamp at a rate of 10 mm/min (Bischetti et al., 2016). Each root segment was pulled using a computer-controlled electronic universal testing machine (CMT6104) with a measuring range of 10 kN and accuracy of 1%. The two ends of each root were wrapped with epoxy resin (Su et al., 2020) to prevent breakage owing to stress concentration during the extension process.

The root diameter was measured at the breaking point (Bischetti et al., 2009) and the root tensile strength was calculated according to the cross-sectional area. Data with the fracture point in the middle of the root were considered valid (Vergani et al., 2016), whereas data with the fracture near the fixed point were deemed invalid. More than 50 sets of valid tests were carried out for each series. A statistical analysis was then performed to obtain the ultimate tensile force and tensile strength of the roots from different burn severity zones and post-fire intervals.

2.4. Soil physical property tests

The bulk density and soil moisture content (SMC) of the undisturbed samples were measured in the laboratory by the ring knife method and oven drying method, respectively, according to United States Department of Agriculture (USDA) standards (USDA, 2004). The fractions of sand (0.05–2 mm), silt (0.002–0.05 mm), and clay (<0.002 mm) were obtained by sieving (Guo et al., 2020) and laser grain size analysis (Mastersizer2000 laser diffractometer). Each group of tests was performed three times to minimize errors (Meftah et al., 2019; Sánchez-Girón et al., 2005).

2.5. Mechanical and hydrological properties tests on soil–root systems

Saturated direct shear tests were performed on the SRS samples obtained from the fire site. Sixteen samples from each burn severity area for all three collection times were sheared in a cylindrical sample holder (diameter = 61.8 mm, height = 20 mm) in a shear testing device (Nanjing Soil Shear Machine SDJ-1, China) after 24 h of vacuum saturation, according to standardized methods (ASTM, 2011). Four vertical pressure values (50, 100, 150, and 200 kPa) and a lateral displacement velocity of 0.8 mm/min of the upper shear box were applied until shear failure occurred. The soil cohesion was obtained as the intercept between the failure line and shear strength axis, and the friction angle was determined from the angle between the failure line and normal stress axis (USDA, 2004; Yao et al., 2022).

Table 2
Category of laboratory tests for remolded and undisturbed soil samples from different burn severity areas.

Condition	Number of samples used in laboratory tests					
	Granulometric composition (R)	Bulk density (U)	SMC (U)	Soil cohesion and friction angle (U)	Hydraulic conductivity (U)	Microstructure observation (U)
Unburned	3	3	3	16	3	3
2 months-low	3	3	3	16	3	3
2 months-medium	3	3	3	16	3	3
2 months-high	3	3	3	16	3	3
1 year-low	3	3	3	16	3	3
1 year-medium	3	3	3	16	3	3
1 year-high	3	3	3	16	3	3
2 years-low	3	3	3	16	3	3
2 years-medium	3	3	3	16	3	3
2 years-high	3	3	3	16	3	3

Note. "R", and "U" refer to remolded soil samples and undisturbed soil samples collected from the site, respectively.

Saturated hydraulic conductivity tests were also carried out on samples obtained from the fire site to determine the hydraulic conductivity of the SRS from areas of different burn severities and post-fire time intervals. Three samples from each burn severity area and post-fire time interval were tested to reduce error, and SRS samples from an unburned area were taken as the control group. Prior to each hydraulic conductivity test, the ring blade method was used to saturate the sample under vacuum. A constant head was used to test the hydraulic conductivity using Darcy's law according to the standard method (ISO, 2004), and the hydraulic conductivity was calculated using:

$$K_s = \frac{qL}{Aht} \tag{1}$$

where K_s is the hydraulic conductivity, q is the infiltration water quantity within the time interval t (5 min), L is the height of the sample (2 cm), A is the cross-sectional area of the sample and held constant at 30 cm^2, and h is the pressure head difference, which was held constant at 100 cm for all tests. The hydraulic conductivity of SRS for each burn severity and post-fire time was estimated as the average of the three hydraulic conductivity test results (Gonzalez-Sosa et al., 2009).

2.6. Microstructure observations of soil–root systems

The degree of root rot and ash cover of the soil surface at different post-fire time intervals were observed by optical microscopy (Nanjing Jiangnan Yongxin NP-800TRF, China) employing 5× objective lenses. Microstructural changes to the SRS samples from different post-fire time intervals were characterized using a Hitachi (S-3000N) scanning electron microscope (SEM, magnification 5–300,000×, acceleration voltage 0–30 kV). The area scanned was 10 × 10 × 3 mm^3 and a magnification of 25× was used. The SRS samples were carefully coated with gold (Mumtaz et al., 2010) to maintain conductivity during scanning (Ahmadi et al., 2021), and three SEM observations for each post-fire interval were conducted to reduce the errors in the results.

2.7. Statistical analyses

Prior to analysis, data were checked for normality and the homogeneity of variances, and log-transformed to correct deviations from these assumptions if necessary. Two-way ANOVA tests were used to assess the combined effects of burn severity and post-fire time on root number, WDPT, soil physical properties, cohesion, internal friction angles, and hydraulic conductivity of SRS. The analyses followed Fisher's least-significant-difference test, considering a significance level of 0.05 (Williams and Abdi, 2010). In addition, the goodness of fit of the relationship between root tensile force and root diameter, as well as the relationship between root tensile strength and root diameter, was evaluated through the coefficient of determination (R^2) and the coefficient of significance with a 95% confidence level. The same operation was applied to assess the relationship between shear strength and vertical pressure.

3. Results

3.1. Post-fire root number and root tensile strength

The variability of root number for each diameter class, burn severity zone, and post-fire time interval were calculated (Fig. 3). The number of roots was substantially different depending on burn severity, post-fire time, and burn severity × post-fire time (Table 3). At each post-fire interval, there were considerably fewer roots in medium-, and high-severity areas than in unburned and low-severity areas. For the fine roots with diameter less than 2 mm, there was almost no difference in root number between the low-severity area and unburned area. While

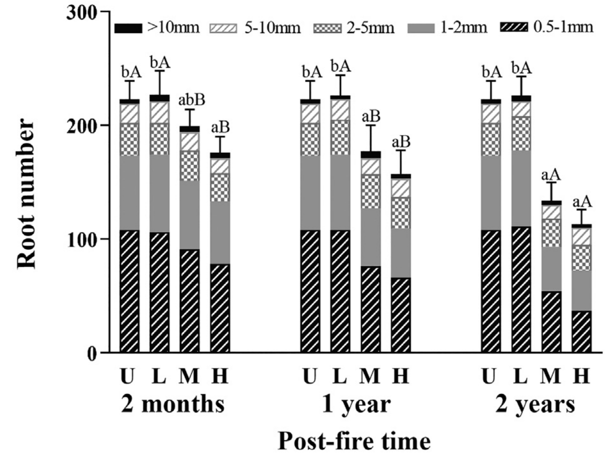

Fig. 3. Root number variations within the cross-sectional area of the trench with time after the forest fire (means (±SD) of three replicates (n = 3)). 0.5–1, 1–2, 2–5, 5–10, and >10 mm refer to root diameter classes, and U (unburned), L (low), M (medium), and H (high) refer to the burn severity. Different lowercase letters indicate significant differences in total number of roots at different levels of burn severity after the same post-fire time interval at $p < 0.05$. Different uppercase letters indicate significant differences in total number of roots at different post-fire time intervals from the same burn severity at $p < 0.05$.

in the medium-severity area, fine roots showed a reduced amount of 13%, 27%, and 46% in the samples collected two months, one year, and two years after the fire, respectively, compared to the control area. Meanwhile, fine roots showed a reduction of 23%, 37%, and 58% respectively in high-severity area. For root number with diameters larger than 2 mm, no obvious reduction was found both in low- and medium-severity area compared with unburned one. For high-severity area, the number of roots decreased by 2%, 6% and 8%, respectively in two months, one year and two years after fire (Fig. 3).

Fig. 4 shows the variations of root ultimate tensile force (F_T) and root tensile strength (T_r) as a function of root diameter (d) at different post-fire time intervals. For the roots sampled at the same time, F_T increased with d and T_r decreased with d following power functions that can be expressed as $F_T = A \cdot d^B$ and $T_r = a \cdot d^b$. The power regression coefficients A, B, a, and b obtained are listed in Table 4. These F_T-d and T_r-d relationships are significant (Table 4) because all the power regression coefficients (A, B, a, and b) and power regressions are highly significant in all cases ($p < 0.05$).

The average T_r values of the roots in different diameter classes in the medium- and high-severity areas were analyzed to better illustrate the variation of T_r with time (Fig. 5). An exponential function was selected for an attenuation model of root strength with time since the wildfire:

Table 3
Results of the two-way ANOVA showing the p-values for root number, WDPT, sand content, silt content, clay content, bulk density, SMC, cohesion, internal friction angle and hydraulic conductivity.

	Burn severity (B)	Df (B)	Post-fire time (T)	Df (T)	B × T	Df (B × T)
Root number	<0.05	3	<0.05	2	<0.05	6
WDPT (s)	<0.05	3	<0.05	2	<0.05	6
Sand content (%)	<0.05	3	0.694	2	0.872	6
Silt content (%)	0.142	3	0.883	2	0.980	6
Clay content (%)	<0.05	3	0.533	2	0.669	6
Bulk density (g/cm^3)	0.490	3	0.902	2	0.999	6
SMC (%)	<0.05	3	<0.05	2	<0.05	6
Cohesion (kPa)	<0.05	3	<0.05	2	<0.05	6
Internal friction angle (°)	0.507	3	0.279	2	0.339	6
Hydraulic conductivity (10^{-5} m/s)	<0.05	3	<0.05	2	<0.05	6

Note. The "Df" refers to the degrees of freedom for each variable.

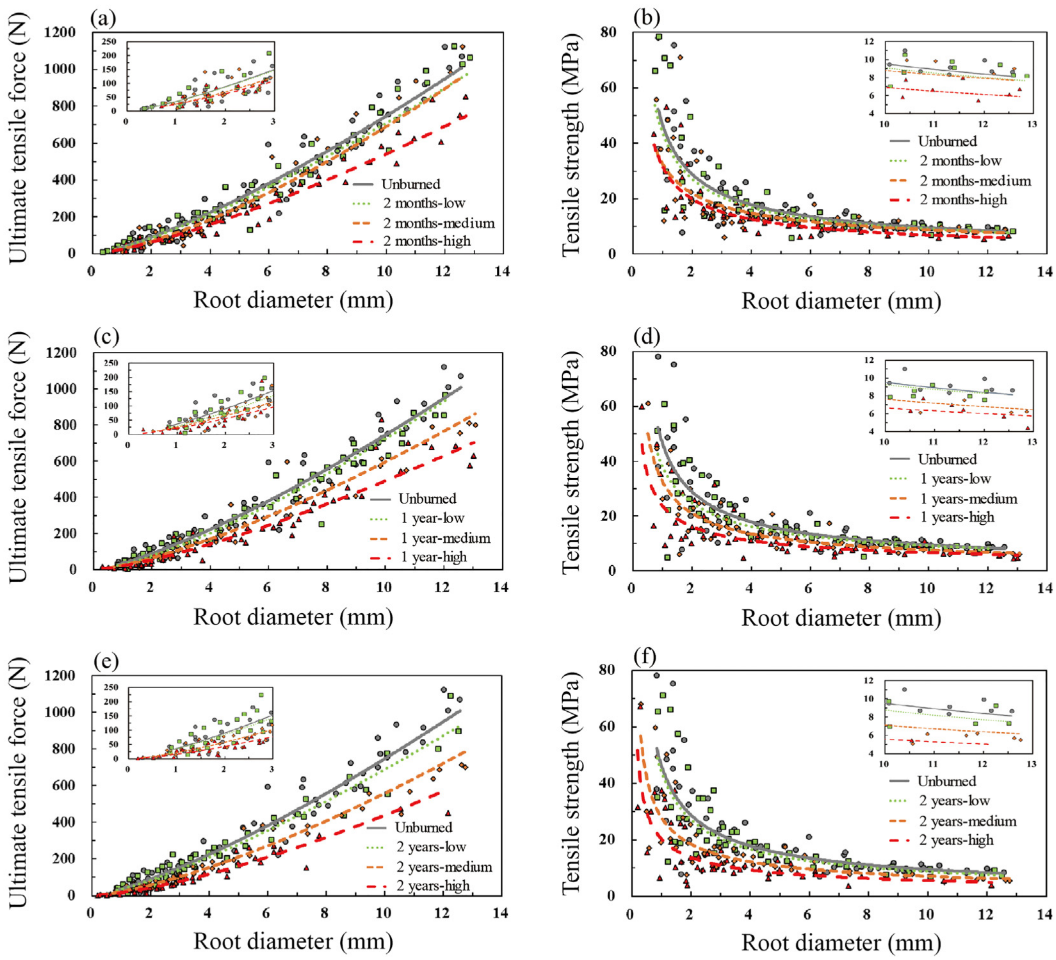

Fig. 4. Relationship between root diameter and (a) (c) (e) ultimate tensile force and (b) (d) (f) tensile strength at different time intervals after the forest fire from areas with different burn severities.

$$T_r(t) = T_0 \cdot e^{-\omega t} \quad (2)$$

where $T_0 = a \cdot d^b$ is the tensile strength of the unburned roots and ω is the attenuation coefficient of the tensile strength, which depends on the root diameter. The slope of the curve represents the decay rate of T_r with time. During the two years after the fire, the decay rate of T_r was larger for roots with $d < 2$ mm, but considerably smaller for roots with $d > 2$ mm (Fig. 5).

3.2. Post-fire soil water repellency and soil physical properties

Soil water repellency after wildfire depends on burn severity and time elapsed since the fire (Table 5). Soil in the unburned area had no water repellency, which was the same for the soil from the low-severity area two months after fire. In contrast, the soil from the medium- and high-severity areas two months after the fire had weak and moderate water repellency, respectively. These results indicate that burn severity has a substantial effect on soil water repellency (Table 3). However, there was no substantial difference in water repellency between soils of various burn severities in one to two years after the forest fire; all samples had no repellency.

The granulometric composition of the soil in the medium- and high-severity areas had changed, whereas no change was observed in the low-severity area (Table 6). In the high-severity area, the sand content of the soil gradually increased from 48.1% ± 4.1% before the fire to 62.3% ± 5.1%, 65.6% ± 3.5%, and 68.1% ± 5.4% two months, one year,

Table 4

Coefficients, *p*-values and coefficients of determination (R^2) for the power regressions between ultimate tensile force F_T and tensile strength T_r with diameter *d*.

Condition	A	p (A)	B	p (B)	R^2	p (F_T–d)	a	p (a)	b	p (b)	R^2	p (Tr-d)	N
Unburned	36.38	<0.01	1.31	<0.01	0.901	<0.01	46.35	<0.01	−0.69	<0.01	0.721	<0.01	62
2 months-low	34.78	<0.01	1.31	<0.01	0.934	<0.01	43.41	<0.01	−0.68	<0.01	0.743	<0.01	56
2 months-medium	25.49	<0.01	1.43	<0.01	0.896	<0.01	32.47	<0.01	−0.57	<0.01	0.670	<0.01	54
2 months-high	24.73	<0.01	1.34	<0.01	0.933	<0.01	31.36	<0.01	−0.66	<0.01	0.652	<0.01	51
1 year-low	29.78	<0.01	1.39	<0.01	0.897	<0.01	37.92	<0.01	−0.61	<0.01	0.644	<0.01	54
1 year-medium	25.52	<0.01	1.37	<0.01	0.853	<0.01	32.47	<0.01	−0.63	<0.01	0.607	<0.01	50
1 year-high	21.81	<0.01	1.35	<0.01	0.832	<0.01	23.71	<0.01	−0.55	<0.01	0.583	<0.01	53
2 years-low	34.64	<0.01	1.30	<0.01	0.894	<0.01	44.13	<0.01	−0.70	<0.01	0.712	<0.01	50
2 years-medium	21.93	<0.01	1.41	<0.01	0.808	<0.01	27.93	<0.01	−0.59	<0.01	0.573	<0.01	52
2 years-high	15.98	<0.01	1.43	<0.01	0.791	<0.01	20.36	<0.01	−0.56	<0.01	0.554	<0.01	50

Note. "2 months", "1 year", and "2 years" refer to the time after the forest fire, and low, medium, and high refer to the burn severity. The uppercase letters "N" refers to the number of data used in regressions.

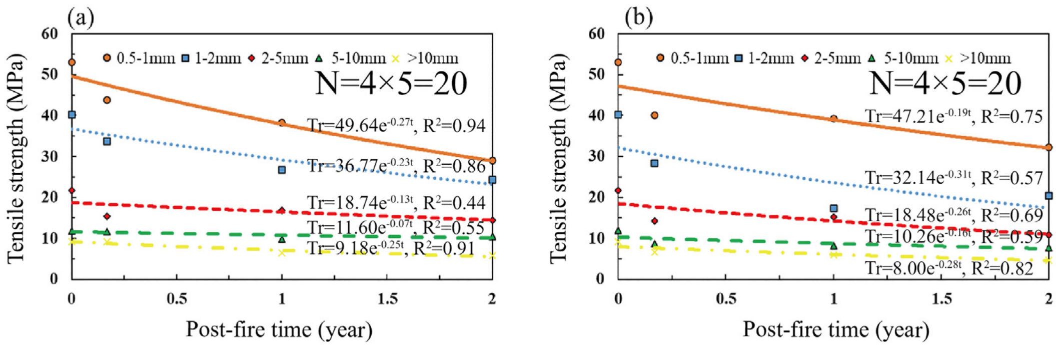

Fig. 5. Variation of the average tensile strength for the different root diameter classes with post-fire time in the (a) medium- and (b) high-burn severity areas.

and two years after the fire, respectively. In contrast, the clay content decreased from 19.3% ± 2.6% to 7.2% ± 1.2%, 6.1% ± 0.5%, and 5.3% ± 0.4% over the same periods. The increased sand content was accompanied by a reduced amount of clay and silt. The bulk density of the unburned soil (1.2 ± 0.0 g/cm^3) was slightly less than the soil two years after the fire (1.3 ± 0.1 g/cm^3, an average value of bulk density of low, medium and high severity burn areas), and there were substantial differences in relation to burn severity while no differences were identified to post-fire time and burn severity × post-fire time (Table 3). The high temperatures also caused a rapid decline of soil moisture two months after the fire, but the soil moisture later returned to pre-fire levels (Table 6).

3.3. Effects of wildfire disturbance on the shear strength of soil–root systems

The shear strength (τ) and normal strength (σ_N) follow a linear relationship (Fig. 6), expressed as $\tau = c + tan\varphi \cdot \sigma_N$, where c and φ are the cohesion and internal friction angles of SRS, respectively. The intercept (c) and slope ($tan\varphi$) of the linear regression is given in Table 7. All the coefficients of c and $tan\varphi$, and the linear regressions were highly significant, with $p < 0.05$.

The responses of the SRS shear strength were sensitive to fire severity and post-fire time interval, as shown in Fig. 6. Under the same normal stress condition, the shear strength of the SRS in the low-severity and unburned areas were similar over the three sampled time intervals. In contrast, the shear strength of the samples in the medium- and high-severity areas decreased substantially after the fire compared with the unburned samples.

Similar to the results discussed in previous sections, the low-severity and unburned areas yielded essentially identical soil cohesion values. The soil cohesion in the medium-severity area decreased by 26%, 37%, and 55% for the samples collected two months, one year, and two years after the fire, respectively. The high-severity area showed cohesion reductions of 48%, 60%, and 82% over the same intervals (Fig. 6),

Table 5
Soil water repellency of different post-fire time and burn severity areas.

Condition	Number of tests	WDPT (s)			
		Min	Max	Mean ± SD	Water repellency
Unburned	10	9.5	16.5	12.1 ± 2.0^{aA}	Non
2 months-low	10	9.1	17.1	12.0 ± 2.5^{aA}	Non
2 months-medium	10	34.3	47.8	40.9 ± 3.8^{bB}	Weak
2 months-high	10	51.6	67.2	61.5 ± 4.6^{cB}	Moderate
1 year-low	10	9.1	16.2	12.7 ± 2.1^{aA}	Non
1 year-medium	10	12.7	16.2	14.4 ± 1.2^{aA}	Non
1 year-high	10	11.4	16.7	13.8 ± 1.8^{aA}	Non
2 years-low	10	8.6	16.5	12.8 ± 2.8^{aA}	Non
2 years-medium	10	10.8	18.2	13.9 ± 2.1^{aA}	Non
2 years-high	10	11.5	15.3	13.3 ± 1.4^{aA}	Non

Note. "2 months", "1 year", and "2 years" refer to the time after the forest fire, and low, medium, and high refer to the burn severity. Different lowercase letters indicate significant differences in WDPT at different levels of burn severity after the same post-fire time interval at $p < 0.05$. Different uppercase letters indicate significant differences in WDPT at different post-fire time intervals from the same burn severity at $p < 0.05$.

Table 6
Physical properties of soil after the forest fire with different burn severities, all the results are the means (±SD) of three replicates ($n = 3$).

Condition	Sand content (%)	Silt content (%)	Clay content (%)	Bulk density (g/cm^3)	Soil texture	SMC (%)
Unburned	48.1 ± 4.1^{aA}	32.6 ± 3.1^{aA}	19.3 ± 2.6^{cA}	1.2 ± 0.0^{aA}	Loam	15.2 ± 1.2^{dA}
2 months-low	50.2 ± 3.8^{aA}	31.3 ± 4.2^{aA}	18.5 ± 2.4^{cA}	1.2 ± 0.0^{aA}	Loam	10.3 ± 0.9^{cA}
2 months-medium	56.4 ± 4.6^{abA}	30.2 ± 2.8^{aA}	13.4 ± 1.8^{bB}	1.2 ± 0.1^{aA}	Sandy loam	7.6 ± 0.5^{bA}
2 months-high	62.3 ± 5.1^{bA}	30.5 ± 2.6^{aA}	7.2 ± 1.2^{aB}	1.3 ± 0.2^{aA}	Sandy loam	4.8 ± 0.7^{aA}
1 year-low	49.2 ± 4.2^{aA}	31.9 ± 2.9^{aA}	18.9 ± 1.5^{cA}	1.2 ± 0.0^{aA}	Loam	14.6 ± 0.3^{aB}
1 year-medium	60.5 ± 4.7^{bA}	28.4 ± 2.3^{aA}	11.1 ± 0.8^{bAB}	1.2 ± 0.1^{aA}	Sandy loam	15.5 ± 1.4^{aB}
1 year-high	65.6 ± 3.5^{bA}	28.3 ± 3.6^{aA}	6.1 ± 0.5^{aAB}	1.3 ± 0.1^{aA}	Sandy loam	14.0 ± 1.2^{aB}
2 years-low	47.1 ± 3.4^{aA}	33.1 ± 4.1^{aA}	19.8 ± 1.3^{cA}	1.2 ± 0.0^{aA}	Loam	15.1 ± 1.5^{aB}
2 years-medium	62.8 ± 4.3^{bA}	27.8 ± 2.7^{aA}	9.4 ± 0.7^{bA}	1.3 ± 0.1^{aA}	Sandy loam	16.2 ± 0.2^{aB}
2 years-high	68.1 ± 5.4^{bA}	26.6 ± 3.8^{aA}	5.3 ± 0.4^{aA}	1.3 ± 0.1^{aA}	Sandy loam	14.7 ± 0.6^{aB}

Note. "2 months", "1 year", and "2 years" refer to the time after the forest fire, and low, medium, and high refer to the burn severity. Different lowercase letters indicate significant differences in physical properties of soil at different levels of burn severity after the same post-fire time interval at $p < 0.05$. Different uppercase letters indicate significant differences in physical properties of soil after different post-fire time intervals from the same burn severity at $p < 0.05$.

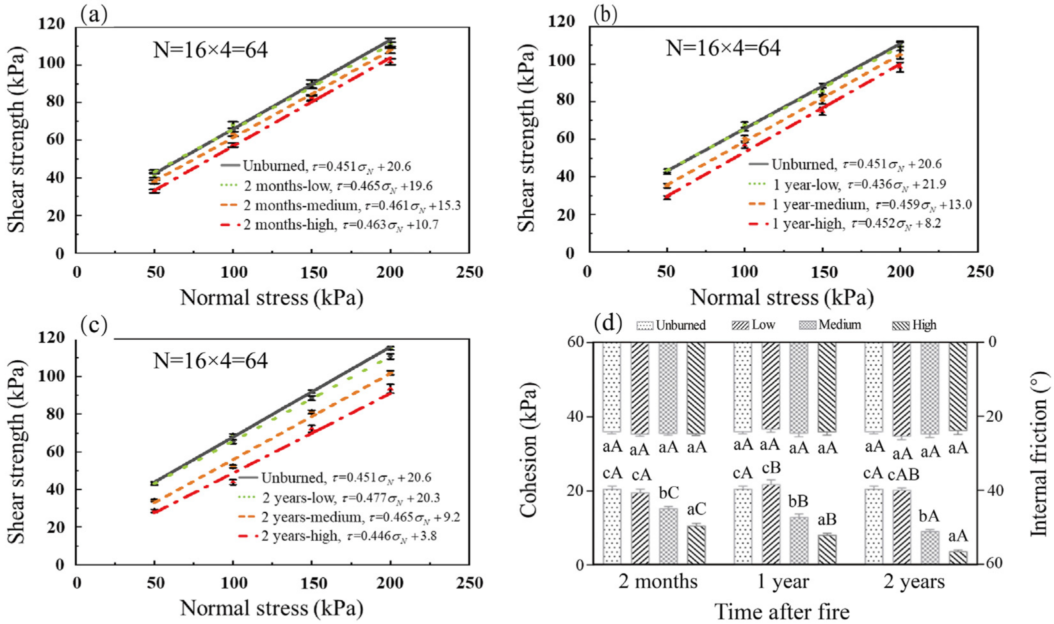

Fig. 6. Relationship between normal stress and shear strength of the soil–root systems (the value of each point represents means (± SD) of four replicates). (a) two months ($N = 16 \times 4 = 64$), (b) one year ($N = 16 \times 4 = 64$), (c) two years ($N = 16 \times 4 = 64$) after the forest fire, and (d) cohesion and internal friction angles of the soil–root systems between the burned and unburned areas at different post-fire intervals. Low, medium, and high indicate the burn severity. Different lowercase letters indicate significant differences in cohesion or internal friction angles of soil at different levels of burn severity after the same post-fire time interval at $p < 0.05$. Different uppercase letters indicate significant differences in cohesion or internal friction angles of soil after different post-fire time intervals from the same burn severity at $p < 0.05$.

which indicates that both burn severity and post-fire time have a substantial influence on cohesion (Table 3). By comparison, the internal friction angles of all the samples showed negligible changes (<2°) with different burn severities and post-fire time intervals.

3.4. Influence of wildfire on the hydraulic conductivity of soil–root systems

The temporal evolution of the saturated hydraulic conductivity (k_s) under different fire severities is shown in Fig. 7. Both burn severity and post-fire interval had significant effects on k_s (Table 3). In the early stages after the fire, the k_s value of unburned soil was 1.61×10^{-5} m/s, whereas the values for the low-, medium-, and high-severity areas were 1.55×10^{-5}, 7.85×10^{-6}, and 4.50×10^{-6} m/s, respectively.

Interestingly, the k_s values of the medium- and high-severity areas increased significantly one to two years after the fire and ultimately exceeded the k_s of the unburned area (Fig. 7). For example, the k_s values of the medium- and high-severity areas increased to 2.62×10^{-5} and 3.24×10^{-5} m/s, respectively, after one year and further increased to 3.52×10^{-5} and 5.15×10^{-5} m/s after two years.

Table 7
Coefficients, *p*-values, and coefficients of determination (R^2) for the linear regression between normal stress and shear strength of the soil–root systems.

Condition	c	p (c)	$tan\varphi$	p ($tan\varphi$)	R^2	p ($\tau - \sigma_N$)	N
Unburned	20.6	<0.01	0.451	<0.01	1.000	<0.01	4
2 months-low	19.6	0.014	0.465	<0.01	0.996	<0.01	4
2 months-medium	15.3	0.023	0.461	<0.01	0.996	<0.01	4
2 months-high	10.7	0.047	0.463	<0.01	0.996	<0.01	4
1 year-low	21.9	0.013	0.436	<0.01	0.995	<0.01	4
1 year-medium	13.0	0.019	0.459	<0.01	0.997	<0.01	4
1 year-high	8.2	0.048	0.452	<0.01	0.990	<0.01	4
2 years-low	20.3	<0.01	0.477	<0.01	0.999	<0.01	4
2 years-medium	9.2	0.039	0.465	<0.01	0.998	<0.01	4
2 years-high	3.8	0.021	0.446	<0.01	0.983	<0.01	4

Note. "2 months", "1 year", and "2 years" refer to the time after the forest fire, and low, medium, and high refer to the burn severity. The uppercase letters "N" refers to the number of data used in regressions.

3.5. Microscopic features of soil–root systems after different post-fire time

The ash coverage, degree of root decay, and soil microstructure of the SRS varied greatly after the fire (Fig. 8). For the unburned SRS, the root-reinforced soil particles were cemented by organic matter and formed a large number of soil aggregates (Fig. 8a). For the samples collected two months after the fire, the soil was clearly darkened from the addition of ash/carbon, and the soil skeleton was filled with a large amount of ash (Fig. 8b). The black ash on the soil surface had disappeared from the SRS samples collected one year after the fire, while gaps had formed from the rotted roots (Fig. 8c). The roots had almost completely decayed and hollowed out from the SRS samples collected

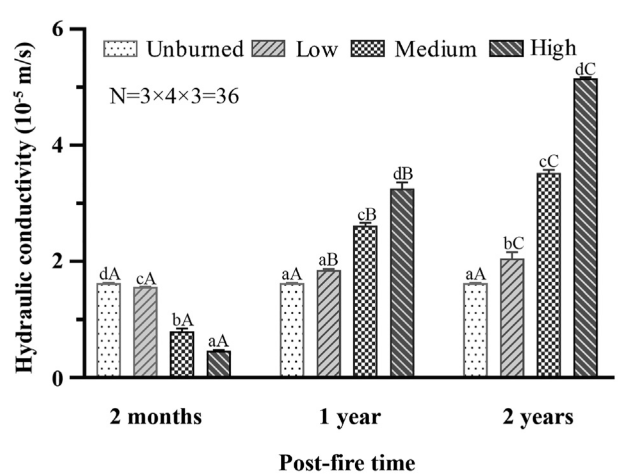

Fig. 7. Variations of the hydraulic conductivity (means (±SD) of three replicates ($n = 3$)) with post-fire time in areas of different burn severities ($N = 3 \times 4 \times 3 = 36$). Different lowercase letters indicate significant differences in hydraulic conductivity of soil at different levels of burn severity after the same post-fire time interval at $p < 0.05$. Different uppercase letters indicate significant differences in hydraulic conductivity of soil after different post-fire time intervals from the same burn severity at $p < 0.05$.

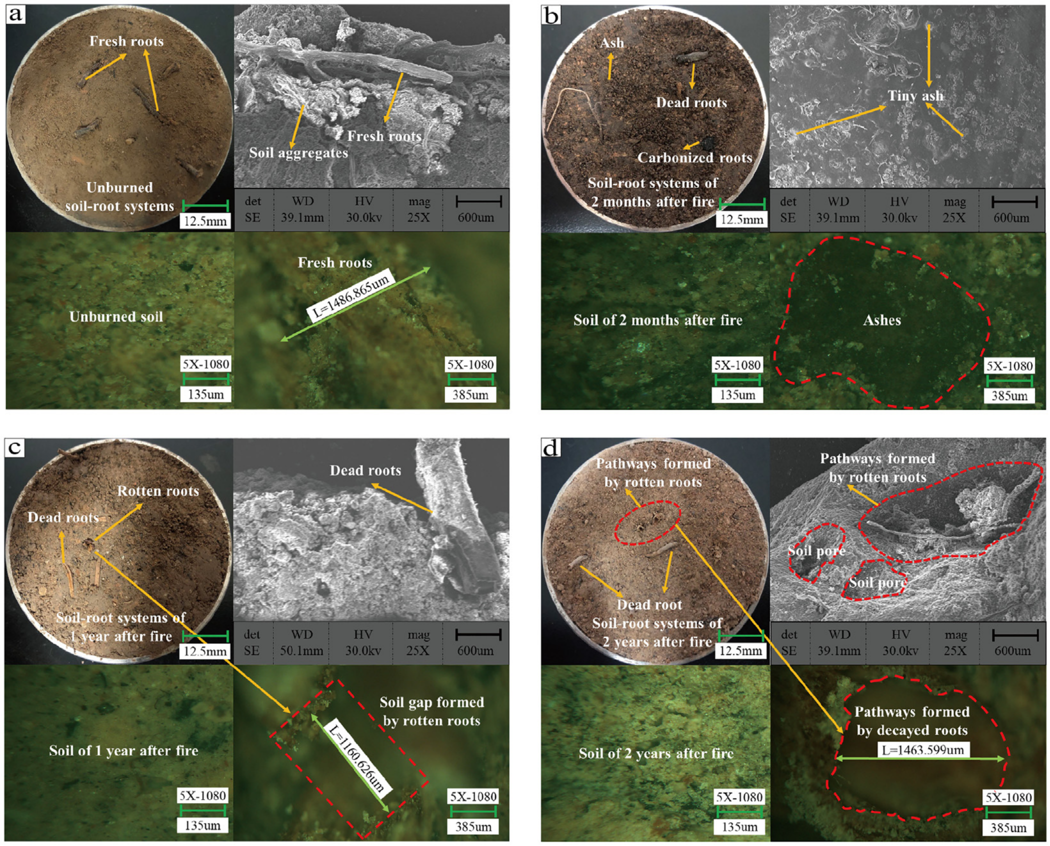

Fig. 8. Surface microscopic images and microstructure of the soil–root systems at different post-fire times: (a) unburned and (b) two months, (c) one year, and (d) two years after the fire.

two years after fire. Only the skins of the roots were left in the soil, which results in the formation of pathways in the soil (Fig. 8d).

4. Discussion

4.1. Mechanical characteristics of soil–root systems subjected to wildfire

Previous studies have established a power function reduction relationship between T_r and d in healthy roots (De Baets et al., 2008; Schwarz et al., 2013). This relationship remains valid for tree roots subjected to fire, although the mechanical properties of the roots gradually decrease with post-fire time (Fig. 4). Previous studies have shown that cellulose content has a significantly positive correlation with T_r (Löbmann et al., 2020) and that decreased cellulose content in rotten roots (Wang et al., 2014) may reduce the T_r of decayed roots with time after a fire. It should be noted that the low-severity fire had little effect on the roots, mainly because low soil temperatures rarely affect the root system below the ground (Heath et al., 2015). The fitting degree (R^2) of the relationship between T_r and d is found to gradually decrease with time after the fire (Table 4). This might be a result of the increased amount of rotten roots, which introduces greater uncertainties and deviations in the measurements of the mechanical properties of roots.

The general trends of exponential decline of average T_r values observed in this study indicate that fire can cause root death, and hence deteriorate the mechanical properties of the roots. The average T_r values determined here for roots finer than 2 mm decreased by 36% (medium severity) to 47% (high severity) two years after the fire. This might be because of the larger specific surface area of fine roots, which are thus more likely to be decomposed by microorganisms in the soil after fire damage (Vergani et al., 2017b), while thicker roots require a longer time to decay. Silver and Miya (2001) found that coarse roots decompose more slowly than fine roots, which may explain the more notable decline of the mechanical strength of the latter.

The important role of vegetation in stabilizing hillsides is well known (Cui et al., 2012; Hales and Miniat, 2017; Pallewattha et al., 2019; Zhang et al., 2020). Living plants can increase the shear strength of soil via mechanical effects and evapotranspiration-induced suction (Ng et al., 2020; Ni et al., 2018; Ni et al., 2019). Here, we found that plant roots tend to increase the shear strength of soil primarily by increasing the soil cohesion, which is consistent with the mainstream understanding (Wu, 2013). The contribution of roots to the soil strength depends on several factors, including root number, distribution, and mechanical properties (Bordoni et al., 2020; Vergani et al., 2017a; Veylon et al., 2015). Our results show that fire can burn roots and that dead roots will gradually decay with increasing post-fire time. This process reduces the root number and degrades their mechanical properties, which can explain the rapid decline of cohesion of the SRS with time after the fire (Fig. 6).

In addition, the sand content of the soil increased by 14.7% (medium severity) and 20.0% (high severity) in the two years following the fire, whereas the clay content decreased by 9.9% (medium severity) and 14.0% (high severity) (Table 6). This observation is attributed to the release of alumina and hydroxide during clay decomposition under high temperatures. These acted as binders in the formation of sand-sized particles (Terefe et al., 2008), causing finer clay particles to aggregate into stable sand-sized particles, thus increase the average particle size (Granged et al., 2011; Ulery and Graham, 1993). Soil erosion processes after a fire can also selectively remove finer particles (e.g., clay), which results in a relative increase in the amount of coarser particles (Granged et al., 2011; Mermut et al., 1997). This series of changes can also lead to a reduction of soil cohesion (Khayat and Alboshoke, 2014). In summary, the cohesion of the SRS after a fire decreases over time because of decreased number, degraded mechanical properties of remaining roots, and the gradual decrease of clay content.

4.2. Hydrological characteristics of soil–root systems after a wildfire

In the early stage after the fire, the hydraulic conductivity decreased to 49% (medium-severity area) and 28% (high-severity area) of that in the unburned area. There are several mechanisms that can reduce the hydraulic conductivity of soil, such as ash clogging (Woods and Balfour, 2010), soil sealing (Larsen et al., 2009), hydrophobicity (Doerr et al., 2000), hyper dry conditions (Moody and Ebel, 2012), loss of ground cover (Morris and Moses, 1987), and changes in macropore flow paths (Nyman et al., 2014). According to existing research, the moisture content of hyper-dry soil is about 0.6% (Moody and Ebel, 2012). The lowest soil water content observed in our samples was 4.8%, which is an order of magnitude higher than that of hyper-dry soil. Through field investigations, we observed that the fire had caused the increase of litter, which increased the coverage of the soil surface. Moreover, the surface of the burned area was loose, and there were no indications of soil sealing or decrease of macropore flow. Based on these observations and the results of the WDPT tests (Table 5) and scanning electron microscopy (Fig. 8b), the causes of the observed reduction of hydraulic conductivity were ash clogging and hydrophobicity.

The WDPT tests showed that the soil from the medium- and high-severity areas transitioned from non-water repellency to weak and moderate water repellency, respectively, in the two months after wildfire (Table 5). This was a result of water-repellent material which formed from the burning of soil organic matter and plant roots (Finley and Glenn, 2010; Mao et al., 2019). This organic matter coated the soil particles as a gas and was eventually deposited on the particle surface. This material is generally hydrophobic and led to extreme water repellency of the soil (Cawson et al., 2012; Doerr et al., 2000; Vieira et al., 2015).

Direct evidence from scanning electron microscopy observations (Fig. 8b) demonstrated that ashes clogged pores of the soil structure, which supports the view that pore clogging caused the decrease in hydraulic conductivity (Pereira et al., 2013; Varela et al., 2015; Woods and Balfour, 2008; Woods and Balfour, 2010). A recent study, however, presented a completely opposite view. Stoof et al. (2016) added ashes to pure sand and observed the migration of the ashes with a microscope. They found that the ashes did not clog soil pores. The two contradictory results may be the result of a pore size threshold, between sandy loam and loamy sand, that determines whether or not ash clogging will occur (Stoof et al., 2016). The loam soil studied by Woods and Balfour (2010) and collected for our study was much finer than the pure sand used by Stoof et al. (2016). Therefore, it was easier to observe ash clogging under the experimental conditions of this study.

Moreover, a significant increase in the hydraulic conductivity was observed for soil from medium- and high-severity zones one to two years after wildfire. This was associated with the disappearance of hydrophobicity caused by the fire in the same period (Blank et al., 1995; Cerdà, 1998; Lucas-Borja et al., 2018; Pereira et al., 2013b; Pereira et al., 2014; Robichaud et al., 2008). The reduced hydrophobicity may be caused by rainwater erosion, which can wash away the ash and hydrophobic compounds in the soil (Granged et al., 2011; Pereira et al., 2013). Furthermore, the pathways formed by rotten roots (Fig. 8c–d) could create an interface priority flow that could increases the SRS hydraulic conductivity. Previous studies have shown that the hydraulic conductivity of soil can increase by a factor of 1.3–6.5 because of root decay or aging (Ni et al., 2017; Vergani and Graf, 2016). This explains why the hydraulic conductivity of the soil in this study increased by factors of 2.2 and 3.2 in the medium- and high-severity areas, respectively, two years after the fire compared with the soil in the unburned area.

It must be noted that the positive head associated with the hydraulic conductivity tests in this study could overwhelm repellency (Ebel, 2019; Nyman et al., 2014). However, only soil from the medium- and high-severity areas was hydrophobic, while the rest of the samples were non-repellent (Table 5). This indicates that water repellency was not the only factor reducing infiltration. Indeed, ash clogging and subsequent root decay also played a key role in controlling the hydraulic conductivity. Previous studies have indicated that a constant head penetration test can be used when ash clogs pores or macropore flow dominates (Stolte, 1997; Stoof et al., 2016), which support the feasibility of the hydraulic conductivity tests we performed. In addition, three SRS samples from each condition were tested to reduce the error resulting from constant head penetration tests, with the average value taken as the final result.

Last but not least, this study assumed that the soil properties of the control group were constant over time. However, for the characteristics of SMC and hydrophobicity that seriously depend on precipitation-evapotranspiration regime which might vary with time (Ferreira et al., 2016; Jiménez-Pinilla et al., 2016), this assumption can cause errors to the test data related to soil hydraulics. In order to reduce the possibility of error caused by this assumption, the sampling points during each period were located at the same location, and the sampling time was relatively close to the previous year to reduce the interference of external environmental conditions on the hydraulic properties of soil (Robichaud et al., 2013b; Vergani et al., 2016). It can be found that the soil showed hydrophobicity only in the medium- and high-severity areas two months after fire, while it did not show water repellency one and two years after fire (Table 5). Further, combined with Table 6, it can be found that the SMC changed greatly two months after the fire, while for the soil of one and two years after the fire, the SMC was close to that of the control group. This phenomenon can prove that the hydraulic conditions among control group and samples from different burn severity (unburned, low-, medium-, and high-severity) areas were similar during that time. Based on the above two analysis, it is reasonable to ignore the natural variability across time in the control group (unburned area). In addition, most of the tests in this study have been tested for three replicates, which can relatively reduce the error caused by environmental factors (Gonzalez-Sosa et al., 2009; Meftah et al., 2019; Sánchez-Girón et al., 2005). We believe that more sufficient test replicates should be considered in future research, which will make the research results more objective.

4.3. Temporal evolution of failure mechanisms for soil–root systems

The destruction of SRS after a fire is the result of cumulative effects and interactions of the mechanical and hydraulic properties of SRS. The process also evolves over time because of the death and decay of roots as well as changes in soil structure. Insight into the state of SRS at different post-fire intervals helps improve the understanding of failure mechanisms of SRS after a fire.

We have found that abundant, healthy roots with high tensile strength can effectively reinforce soil (Fig. 6) in unburned areas, consistent with existing studies (Giadrossich et al., 2013; Löbmann et al., 2020; Schwarz et al., 2010), and illustrated in Fig. 9(a).

In the early stage after fire (two months after fire in this study), roots that had been burned by the high temperatures of the fire began to die, which resulted in a decrease in root tensile strength (Fig. 5). Soil structure was also damaged and filled with a large amount of ash (Fig. 8b) and hydrophobic compounds, which can lead to a decrease of shear strength and hydraulic conductivity of SRS. This series of changes will typically result in increased sheet erosion and slope erosion in a burned area, as observed during field investigations (Fig. 1(g) and (h)), and consistent with the observation by Wall et al. (2020). The state of SRS in this period is illustrated in Fig. 9(b).

One year after wildfire, the decay of roots had resulted in a further decline of the saturated shear strength of SRS. Additionally, the disappearance of soil hydrophobicity (Table 5) and the development of soil gaps formed by rotten roots (Fig. 8c) would lead to an increase in hydraulic conductivity (Fig. 7), which would allow rain to infiltrate deep into the soil (Fig. 9c) (Wei et al., 2021). Two years after the fire, the formation of pathways from the continuous decay of dead roots further increases the hydraulic conductivity of soil. This means that rainwater can

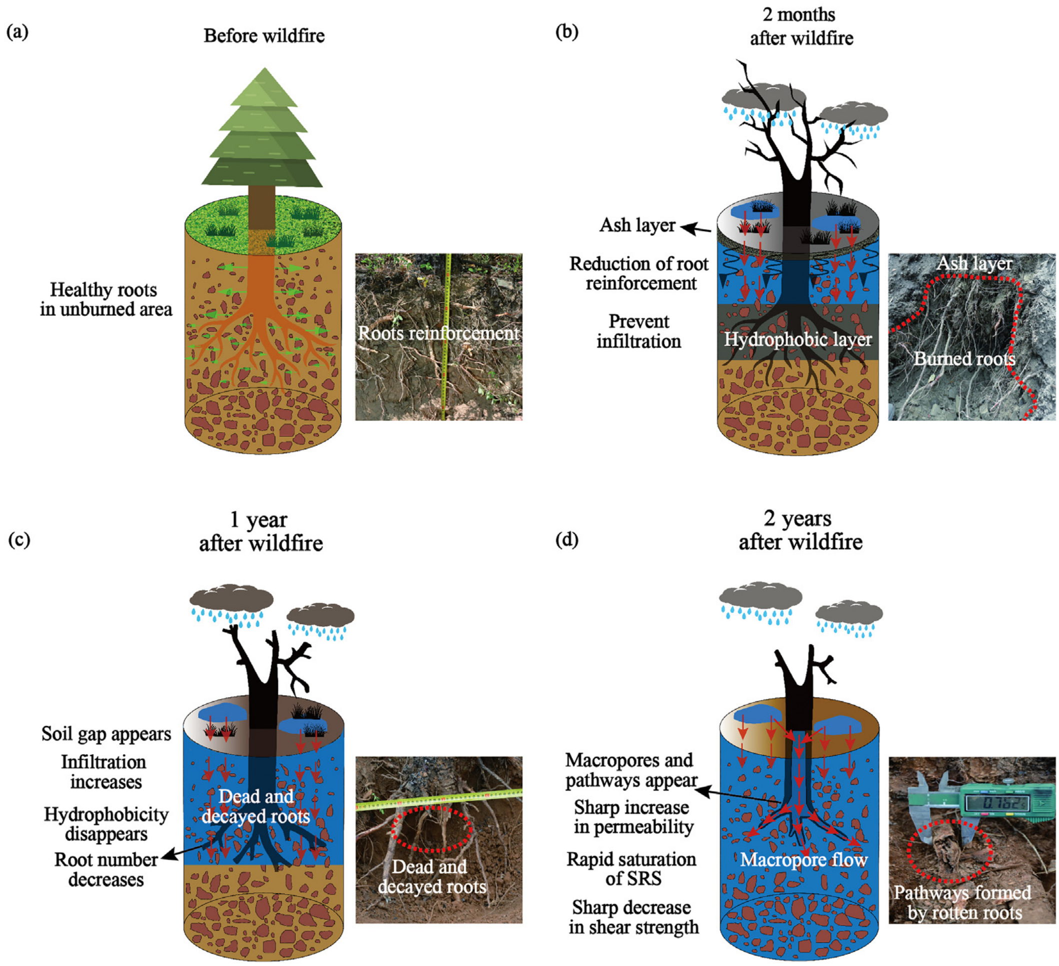

Fig. 9. Temporal evolution of the failure mechanisms for soil–root systems after a fire. (a) Prior to the fire, the roots strengthen the soil; (b) two months after fire, ash from burned vegetation covers the soil surface and clogs soil pores. Soil hydrophobicity increases and hydraulic conductivity decreases; (c) one year after the fire, soil hydrophobicity disappears and hydraulic conductivity increases; (d) two years after the fire, root reinforcement of the soil decreases and macropores and pathways appear because of root decay.

more easily enter the soil via macropore flow (Cui et al., 2017; Jiang et al., 2018; Jørgensen et al., 2019) and accelerate the saturation of the SRS (Fig. 9d). Moreover, the decayed roots result in the nearly complete decline of the SRS cohesion (Fig. 6), thus allowing soil failure to readily occur. Indeed, the field investigations for this study found that shallow landslides had appeared (Fig. 1i) two years after the fire in the study area; similar results were previously reported by Meyer et al. (2001) and Rengers et al. (2020).

The mechanisms of SRS failure after a fire are closely related to the deterioration of SRS hydro-mechanical properties. Some specific measures can improve the hydraulic and mechanical properties after fire. The spread of organic material (e.g., wheat straw, wood products, mulch) has been shown to improve the mechanical and hydrological properties of soil and substantially reduce runoff and soil erosion (Pereira et al., 2018). These materials are thus being increasingly applied as a restoration treatment after a wildfire (Lucas-Borja et al., 2020; Lucas-Borja et al., 2018; Robichaud et al., 2013a).

5. Conclusions

After two consecutive years of research, results demonstrated significant degradation of the hydro-mechanical properties of SRS as a consequence of the wildfire. Two years after the wildfire, root number and root tensile strength declined seriously because of root death and subsequent decay after the fire, resulting in reduction of the SRS cohesion. The saturated hydraulic conductivity of the SRS from medium- and high-severity area decreased in the two months after fire, as a result of ash clogging and the formation of a hydrophobic layer. However, the disappearance of the hydrophobic layer and the formation of macropore flow pathways formed by rotten roots finally increased the soil hydraulic conductivity to 2.2–3.2 times higher than the pre-fire level two years after fire.

These findings provide important insight on the time-dependent evolution of the hydro-mechanical properties of SRS after fires and are critical for understanding the initiation mechanism of post-fire debris flows and shallow landslides. Caution should be taken that the results of this study are from only one small-scale wildfire in a subalpine conifer forested area. Similar issues and research about change laws of SRS hydromechanical properties with burn severity and post-fire time should be carried out in other parts of the globe.

CRediT authorship contribution statement

Mingyu Lei: Conceptualization, Methodology, Investigation, Validation, Data curation. **Yifei Cui**: Conceptualization, Validation, Supervision, Funding acquisition, Writing-review & editing, Project administration, Investigation. **Junjun Ni**: Conceptualization, Validation, Supervision, Project administration, Writing-review & editing, Investigation. **Guotao Zhang**: Validation, Writing-review & editing, Investigation, Supervision, Formal analysis. **Yao Li**: Methodology, Investigation, Formal analysis, Writing-review & editing. **Hao Wang**: Investigation, Review & editing. **Dingzhu Liu**: Review & editing. **Shujian Yi**: Methodology, Review. **Wen Jin**: Review. **Liqin Zhou**: Review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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